



Zhenghao Zhang

📍 Munich & Stuttgart, Germany 🗓 Dec. 1999
🌐 github.com/zhenghao2519 🏠 <https://zhenghao2519.github.io/>
📧 [zhenghao2519](mailto:zhenghao2519@tum.de) 📞 +49 15154157038 @ zhenghao.zhang@tum.de

🎓 Education

Sep. 2026	School of CIT, TU Munich	
Oct. 2023	Master in Informatics	Grade: 1.5 / 4.0
Sep. 2023	Faculty of Computer Science, RWTH Aachen	
Oct. 2019	Bachelor in Informatics	Grade: 2.2 / 4.0

Scale: 1.0 - excellent, 4.0 - sufficient

📄 Publications

- **SpaceDrive: Infusing spatial awareness into VLM-based autonomous driving**
Peizheng Li*, Zhenghao Zhang*, David Holtz, Hang Yu, Yutong Yang, Yuzhi Lai, Rui Song, Andreas Geiger, Andreas Zell
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- **Self-sovereign Identity-based Access Control Management in Forestry 4.0**
Yongli Mou*, Jiahang Chen*, Zhenghao Zhang*, Jürgen Roßmann, Stefan Decker
International Conference on Future Internet of Things and Cloud (FiCloud), 2023

🔗 Projects

- **Beyond-Fixed-Forms: Query-aware 3D Instance Segmentation**
Proposed a plug-and-play open-vocabulary model optimization pipeline, utilizing VLMs target re-discovery and refining fixed-shape 3D masks to align with language inputs. Achieved significant performance improvement (~20%↑) over SOTA on the ScanNet200 dataset. Code released at: *Beyond-Fixed-Forms*.
- **OrganizAR: Medical AR Tool for Operating Room Preparation**
Generated instructions based on robust scene understanding powered by a VLM-driven open-vocabulary model. Specifically designed mask fusion and filtering algorithms for complex surgical environments, enabling precise localization of rare medical instruments. Code: *OrganizAR*.
- **Decentralized Access Control in Forestry 4.0**
Explored a decentralized-identity-based access control management system for heterogeneous IoT devices in Forestry 4.0. The project is partially funded by a funding program from the NRW government.

👛 Experience

Sep. 2026	Internship, Working Student & Master Thesis @ Mercedes-Benz AG
Apr. 2025	➤ Use positional encodings to inject 3D spatial awareness into Vision-Language Model (VLM)-based autonomous driving systems. ➤ Achieved 1st place on the open-loop nuScenes benchmark and 2nd place among VLM approaches on the closed-loop Bench2Drive benchmark. The paper is accepted by CVPR 2026 and open-sourced at <i>SpaceDrive</i>.
Oct. 2025	Interdisciplinary Project @ fortiss
Nov. 2024	➤ Developed a multi-modal fusion framework integrating point cloud and camera inputs via QFormer to enhance downstream VLM-based autonomous driving.
Apr. 2025	Research Internship @ Helmholtz Munich
Oct. 2024	➤ Implemented an adaptive PEFT system utilizing Multi-Expert LoRA with a routing mechanism to boost few-shot and incremental learning capabilities.
Apr. 2025	Student Assistant at TU Munich
Jul. 2024	➤ Introduction to Deep Learning by Prof. Matthias Nießner (2024SS) and Prof. Daniel Cremers. (2024/25WS).

🔧 Competences & Languages

Programming	Python, PyTorch, Shell, C++; MMCV, MMDet3D; Git, Docker, Slurm.
Office & Writing	LaTeX, Markdown, Microsoft Office
🌐 Languages	English (IELTS 7.0), German (C1), Mandarin (native)